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LOW-PROFILE MODULAR POSITIONING COUPLER

Abstract

A modular coupler coaxially mountable with a mast includes a first plate, a second plate and a column axially extending between the first and second plates. An arm is rotatably attached to the column and generally transversely extends therefrom and beyond a radial extent of the first plate and of the second plate. A mounting bracket projects from the arm along a portion of the arm outside the radial extent of the first plate and of the second plate. The mounting bracket is configured to removably mount a first surveillance equipment thereto. A motor is mounted to the arm outside the radial extent of the first plate and of the second plate and is configured to rotate the arm about the column.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of similarly-titled, co-pending U.S. patent application Ser. No. 18/635,239, filed on Apr. 15, 2024, which claims priority to U.S. patent application Ser. No. 17/895,133, filed on Aug. 25, 2022, and issued as U.S. Pat. No. 11,988,326 on May 21, 2024, which claims priority from similarly-titled U.S. Provisional Patent Application No. 63/237,863, filed Aug. 27, 2021, the entire contents of each of which are incorporated by reference herein in their entireties.

BACKGROUND OF THE DISCLOSURE

[0002] The disclosure generally relates to surveillance systems, and more particularly, to a low-profile, modular positioning coupler configured for use with surveillance systems.

[0003] Surveillance systems are well-known in the art. One example of such a system is the Freedom-On-The-Move ("FOTM 3.0"), which is a commercially available retractable, vehicle-mounted surveillance system sold by Freedom Surveillance, LLC dba STRONGWATCH®, Scottsdale, Ariz. As shown in FIGS. **1**A and **1**B, the conventional surveillance system **5** is mounted to a bed **3** of a truck **1**. Generally, the system **5** includes a movable, e.g., pneumatically, retractable telescoping structure **6** having a first surveillance equipment **7**, e.g., without limitation, a camera, radar, laser for target designation or range determination, sensor(s), or other payload, removably mounted thereon. In a fully extended, upright, i.e., operational, position (FIG. **1**A), the first retractable telescoping structure **6** effectively functions as a mast and the first surveillance equipment **7** is mounted at or near a top portion of the mast **6**. In a retracted/stowed state, as shown in FIG. **1**B, the system **5** fits entirely within the bed **3** of the truck **1** so that it can be completely hidden and protected from the elements using a covering **2**.

[0004] In addition to the first surveillance equipment **7**, additional surveillance equipment may be beneficial. Additional surveillance equipment may include a camera, radar, sensor(s), flood light, loud hailer, another payload, or a combination thereof. Similarly to the first surveillance equipment **7**, it is desirable for the additional surveillance equipment to be retractable when not in use, e.g., to be hidden from view and protected from the elements, and which can be deployed during use. One drawback of conventional mobile surveillance systems is that a second retractable structure (not shown), operating as a second mast, is employed, to mount the additional surveillance equipment thereof. Alternatively, in some systems, a cross-bar (not-shown) is mounted to the first retractable telescoping structure **6**, and both the first surveillance equipment **7** and the additional surveillance equipment are mounted side-by-side along the cross-bar.

[0005] Both of such approaches result in a much larger system, that is heavier, more expensive, and more cumbersome to manufacture and operate. Additionally, such systems result in operational deficiencies. For example, in the cross-bar formation, the surveillance devices may interfere with the line of sight/field of regard of one another. Additionally, any surveillance devices that are offset from the central axis of the telescoping structure **6** when mounted create a bending force on the telescoping structure **6**, thereby subjecting the telescoping structure **6** to unnecessary forces which may interfere with mobility of the system and/or otherwise amplify the risk of telescoping structure **6** failure.

[0006] It would, therefore, be advantageous to manufacture a modular, low-profile, positioning coupler mountable coaxially with the retractable, telescoping structure for mounting of additional equipment surveillance thereto.

BRIEF SUMMARY OF THE DISCLOSURE

[0007] Briefly stated, one aspect of the present disclosure is directed to a modular coupler coaxially mountable with a mast. The modular coupler includes a first plate, a second plate and a column axially extending between the first and second plates. An arm is rotatably attached to the column and generally transversely extends therefrom and beyond a radial extent of the first plate and of the second plate. A mounting bracket projects from the arm along a portion of the arm outside the radial extent of the first plate and of the second plate. The mounting bracket is configured to removably mount a first surveillance equipment thereto. A motor is mounted to the arm outside the radial extent of the first plate and of the second plate and is configured to rotate the arm about the column.

[0008] Another aspect of the present disclosure is directed to a modular coupler coaxially mountable with a mast. The modular coupler includes a spool-shaped frame having a first plate, a second plate, and a column axially extending between the first and second plates. An arm is rotatably attached to the column and generally transversely extends therefrom and beyond a radial extent of the first plate and of the second plate. A mounting bracket projects from the arm along a portion of the arm outside the radial extent of the first plate and of the second plate. The mounting bracket is oriented substantially parallel with the column and is configured to removably mount a first surveillance equipment thereto. A motor is mounted to the arm outside the radial extent of the first plate and of the second plate and is configured to rotate the arm about the column. A stationary toothed pulley is fixedly mounted about the column and a rotatable drive pulley, operatively connected to the motor, is positioned outside the radial extent of the first plate and of the second plate. A toothed belt extends between the rotatable drive pulley and the stationary toothed pulley. The second plate is configured to mate with a base mounting plate of a second surveillance equipment, the second plate including a clamp configured to removably engage the base mounting plate of the second surveillance equipment in a tool-free manner, whereby the second surveillance equipment is coaxially mountable upon the coupler.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] The following description of the disclosure will be better understood when read in conjunction with the appended drawings. It should be understood, however, that the disclosure is not limited to the precise arrangements and instrumentalities shown. In the drawings:

[0010] FIG. **1**A is a perspective view of a conventional mobile surveillance system in an operational state;

[0011] FIG. **1**B is a perspective view of the conventional mobile surveillance system of FIG. **1**A in a stowed away state;

[0012] FIG. **2** is an elevational view of a positioning coupler of the present disclosure;

[0013] FIG. **3** is a cross-sectional view of the positioning coupler of FIG. **2**;

[0014] FIG. **4** is an enlarged, partial view of a motor and arm of the positioning coupler of FIG. **2**;

[0015] FIG. **5** is an elevational view of the positioning coupler of FIG. **2** mounted to the mobile surveillance system of FIG. **1**; and

[0016] FIG. **6** is a perspective view of the positioning coupler of FIG. **2** mounted to the mobile surveillance system of FIG. **1**.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0017] Certain terminology is used in the following description for convenience only and is not limiting. The words “lower,” “bottom,” “upper” and “top” designate directions in the drawings to which reference is made. The words “inwardly,” “outwardly,” “upwardly” and “downwardly” refer to directions toward and away from, respectively, the geometric center of the positioning coupler, and designated parts thereof, in accordance with the present disclosure. In describing the positioning coupler, the term proximal is used in relation to the upper end of the device and the term distal is used in relation to the bottom end of the device. Unless specifically set forth herein, the terms “a,” “an” and “the” are not limited to one element, but instead should be read as meaning “at least one.” The terminology includes the words noted above, derivatives thereof and words of similar import.

[0018] It should also be understood that the terms “about,” “approximately,” “generally,” “substantially” and like terms, used herein when referring to a dimension or characteristic of a component of the disclosure, indicate that the described dimension/characteristic is not a strict boundary or parameter and does not exclude minor variations therefrom that are functionally similar. At a minimum, such references that include a numerical parameter would include variations that, using mathematical and industrial principles accepted in the art (e.g., rounding, measurement or other systematic errors, manufacturing tolerances, etc.), would not vary the least significant digit.

[0019] Referring to the drawings in detail, wherein like numerals indicate like elements throughout, there is shown in FIGS. **2-6**, a positioning coupler, generally designated **10**, in accordance with an embodiment of the present disclosure for mounting to a retractable, telescoping structure **6**, e.g., a retractable telescoping structure of the surveillance system **5**, as described in further detail below.

[0020] In the illustrated embodiment, the positioning coupler **10** takes the form of a spool-shaped coupler. That is, the positioning coupler **10** includes a bottom (first) plate **14** and a top (second) plate **12**. As shown best in FIG. **3**, the bottom plate **14** includes a first, upwardly projecting throat **14*a***, i.e., in a direction toward the top plate **12**, and the top plate **12** includes a second, downwardly projecting throat **12*a***, i.e., in a direction toward the bottom plate **14**. The first and second throats **14*a***, **12*a*** are configured to fixedly engage one another to form a continuous column **16** extending between the top and bottom plates **12**, **14**. In the illustrated embodiment, the second throat **12*a*** is at least partially received within the first throat **14*a***, but the disclosure is not so limited. For example, the first throat **14*a*** may alternatively be received with the within the second throat **12*a***. Further alternatively, the first and second throats **14*a***, **12*a***, may be otherwise fastened to one another, e.g., without limitation, via engaging faces at the respective terminal ends thereof, to form the continuous column **16**. Yet further alternatively, the column **16** may be formed as a single piece structure projecting from one of the plates **12**, **14** and fastened to the other, or fastened to both plates **12**, **14**.

[0021] An arm **18** is rotatably attached to the column **16** (as will be described in further detail below) and transversely extends therefrom. That is, the arm **18** is rotatable about the column **16**. In the illustrated embodiment, the arm **18** is oriented generally perpendicularly to the column **16**, but the disclosure is not so limited. The length of the arm **18** is greater than the radial extent of the top and bottom plates **12**, **14**. In the illustrated embodiment a terminal, free end of the arm **18** is defined by a mounting bracket **20**. In the illustrated embodiment, the mounting bracket **20** is oriented generally parallel with the column **16**, but the disclosure is not so limited. As should be understood, the mounting bracket **20** may also project from the arm **18** along different portions of the length of the arm **18** and at different angles relative to the arm **18**. As also should be understood, more than one mounting bracket **20** (of the same or differing sizes) may project from the arm **18** along different portions of the length thereof. As shown, surveillance equipment **22** may be removably mounted to the mounting bracket **20** via any of numerous mounting means currently known or that later become known in the art. For example, the surveillance equipment **22** may be fastened to the mounting bracket **20** via fastening screws, bolts, clamps, a combination thereof or the like. In the illustrated embodiment, the surveillance equipment **22** takes the form of a 3D radar, such as, for example, without limitation the EchoGuard 3D Surveillance Radar sold by Echodyne Corp, but the disclosure is not so limited. Additionally, or alternatively, the surveillance equipment

22 may take the form of other radars, sensor(s), camera(s), flood light(s), loud hailer(s), microphone array(s), another payload, or a combination thereof.

[0022] Turning to the column **16**, and as shown best in FIG. **3**, a bearing **28**, e.g., a turntable bearing such as a crossed-roller turntable bearing, is positioned about the column **16**, between the top and bottom plates **12**, **14**. An inner race **28***a* of the bearing **28** is fixedly, i.e., non-rotatably, mounted relative to the column **16** (as will be described in further detail below). Conversely, an outer race **28***b* of the bearing **28** is rotatable about the column **16**. A leading or proximal end **18***a* of the arm **18***a* also encircles the column **16** and is securely attached to the outer race **28***b* of the bearing **28**, thereby enabling the arm **18** to rotate about the column **16** with the outer race **28***b*.

[0023] As shown best in FIGS. **2** and **3**, a toothed pulley **30** is also mounted between the top and bottom plates **12**, **14**, and at least partially encircles the bearing **28** without impacting rotation of the outer race **28***b* thereof. That is the toothed pulley **30** is spaced radially outwardly from outer race **28***b*. The toothed pulley **30** is also fixedly, i.e., non-rotatably, mounted relative to the column **16**. In the illustrated embodiment, and as shown in FIG. **3**, at least one standoff **31** is attached to and projects from the top plate **12**, through the inner race **28***a* of the bearing **28** and into attachment with the pulley **30** to rotationally fix the inner race **28***a* and the pulley **30**, but the disclosure is not so limited. That is, the inner race **28***a* and the pulley **30** may be rotationally fixed relative to column **16** via any of numerous different fastening mechanisms currently known, or that later become known, by those of ordinary skill in the art. At a minimum, the standoff(s) **31** may alternatively be attached to and extend from the inner race **28***a* of the bearing **28**, through the pulley **30** and into attachment with the bottom plate **14**. The standoff(s) **31** also maintains a spacing between the bearing **28** and the pulley **30** relative to the top plate **12** as well as the bottom plate **14**.

[0024] As shown best in FIGS. **2** and **4**, a motor **24** is mounted to the arm **18**. In the illustrated embodiment, the motor **24** is mounted to the underside of the arm **18**, but the disclosure is not so limited. That is, for example, without limitation, the motor **24** may alternatively be positioned above the arm **18** or within the arm **18** (depending on the dimensions of the arm **18**). As should be understood by those of ordinary skill in the art, the motor **24** is operatively connected to a drive pulley **26** which is rotated/spun by the motor **24** in a manner well understood. In the illustrated embodiment, the motor **24** takes the form of a stepper motor, e.g., a NEMA 17-sized motor made by Sanyo Denki of Japan, but the disclosure is not so limited. For example, without limitation, the motor **24** may take the form of a brushless or brushed DC motor, or the like. Gear reduction and/or position sensing may optionally also be employed. A toothed belt **32** extends between the rotating drive pulley **26** and the stationary, toothed pulley **30**. As should be understood, therefore, the teeth of the stationary pulley **30**, engaged with the belt **32**, prevent the belt **32** from spinning. Rather, in operation, rotation of the drive pulley **26** (via operation of the motor **24**) relative to the non-spinning belt **32** rotates the arm **18** (and the outer bearing race **28***b*) about the column **16**. Stated differently, the rotation of the drive pulley **26** pulls itself (and the arm **18**) around the column **16**.

[0025] Advantageously, the motor **24** and the drive pulley **26** are positioned outside the radial extent of the top and bottom plates **12**, **14**. Accordingly, the size of the motor **24** and the drive pulley **26** does not factor into the required distance between the top and bottom plates **12**, **14**. This assists in forming a low-profile positioning coupler **10**. In one embodiment, the top and bottom plates **12**, **14** are only spaced approximately four inches apart, despite the motor **24** and the drive pulley **26** taking up a greater space in the same orientation. Additionally, the surveillance equipment **22** can be substantially larger in the same orientation than the space between the top and bottom plates **12**, **14**. Further advantageously, the rotational ability of the surveillance equipment **22** (due to the rotational ability of the arm **18**) reduces the number of equipment required. For example, panels radars generally scan a region between approximately 90° and approximately 120° without moving the antenna. If the required field of regard is wider than 120°, e.g., 180° or 360°, multiple panels pointed in different directions may be required, which greatly compounds cost as well as weight and sail area aloft, the latter driving a scaling-up of all the supporting equipment (structure and power), perhaps beyond suitability, particularly in cases where an operational requirement is that the entire package of mast and sensors must be stowed in the bed of the truck. Conversely, the positioning coupler **10** reduces the need for duplicative surveillance equipment **22** due to the rotational ability of the arm **18**, enabling substantially 360° range.

[0026] FIGS. **5** and **6** illustrate a non-limiting, exemplary use of the positioning coupler **10** with the system **5**, e.g., the FOTM 3.0. As shown in FIG. **5**, the positioning coupler **10** is mounted between a terminal, upper end of the retractable telescoping structure **6**, i.e., mast, and an underside of the first surveillance equipment **7**. In the illustrated embodiment, the positioning coupler **10** is configured to mount with, and between, the mast **6** and the surveillance equipment **7** via the native mounting structure employed on the FOTM 3.0. That is, the terminal end of the mast **6** includes an upper mounting plate **6***a* configured to removably mount with a base mounting plate **7***a* of the surveillance equipment **7** via a quick connect/disconnect clamp **8** positioned on the upper mounting plate **6***a*. The bottom plate **14** of the positioning coupler **10** is, therefore, formed to mate with the upper mounting plate **6***a* of the mast **6**, and receive the clamp **8** (as previously received by the mounting plate **7***a*) to removably engage the positioning coupler **10** with the mast **6**. The surveillance equipment **7** may then be positioned upon the top plate **12** of the positioning coupler **10**. The top plate **12** of the positioning coupler **10** is formed to mate with the base mounting plate **7***a* of the surveillance equipment **7** in like manner as the upper mounting plate **6***a* of the mast **6**. That is, the top plate **12** includes a clamp **9**, similar to the clamp **8**, to removably engage the base mounting plate **7***a* of the surveillance equipment **7**. Advantageously, utilizing the native mounting structure of the FOTM 3.0 enables plug-and-play use of the positioning coupler **10** with the FOTM 3.0. That is, the system **5** and the surveillance equipment **7** require no additional structural fabrication for use with the positioning coupler **10**. As should be understood by those of ordinary skill in the art, however, the positioning coupler **10** may removably mount between the mast **6** and the surveillance equipment **7** via any of numerous different mounting means currently known or that later become known. Similarly, the mounting bracket **20** can be configured with a quick clamp bracket to facilitate the rapid substitution of different surveillance equipment **22** for use with the positioning coupler **10**.

[0027] Advantageously, the quick clamp mounting structure enables tool free mounting of the positioning coupler **10**. Further advantageously, multiple positioning couplers **10** may be stacked and mounted upon one another to add as many additional surveillance equipment **22** as required. Yet further, the low-profile dimension of the positioning coupler **10**, e.g., approximately four inches between the plates **12**, **14**, adds minimal total length increase to the system **5**. Accordingly, the system **5** is still retractable into the stowed position to fit within the bed **3** of the truck **1**.

[0028] Yet further advantageously, because the first surveillance equipment **7** is stacked above the additional surveillance equipment **22** when the positioning coupler **10** is mounted with the system **5** as previously described, both devices **7**, **22** have a substantially unobstructed field of regard to the horizon. Moreover, because the positioning coupler **10** is mounted coaxially with the underlying mast **6** and the overlying surveillance equipment **7**, the mass of the payload(s) **7**, **22** remain largely aligned with the central axis of the mast **6**. Such a design results in efficient on-the-move operation. That is, there is reduced torque/bending force on the mast **6**, thereby resulting in increased stability and structural integrity of the mast **6** and the overall system **5** during movement of the truck **1**, even in cases when the mast **6** is partially or fully deployed during movement of the truck **1**.

[0029] Turning back to FIG. **3**, the column **16** also includes a lower through-hole **34***a* underlying the bearing **28** and pulley **30** and an upper through hole **34***b* above the bearing **28** and the pulley **30**. The column **16** also defines an internal pathway **16***a* extending between the through-holes **34***a*, **34***b*, thereby enabling communication therebetween. As also shown in FIG. **5**, the system **5** includes power and data cables **36** traveling within a coil **4** wrapped around the mast **6**. The power and data cables **36** exit the coil **4** proximate the upper mounting plate **6***a* of the mast **6**. The power and data cables **36** are then divided into a first power and data cable set **36***a* and a second power

and data cable set **36***b*. As shown, the positioning coupler **10** may include a socket **38**, e.g., a circular connector such as an MIL-DTL-26482 compliant connector, configured to receive the power and data cables **36**, and, thereafter divide into the first power and data cable set **36***a* and the second power and data cable set **36***b*. The first power and data cable set **36***a* is fed underneath the bearing **28** and pulley **30** and into electrical communication with the motor **24** and the surveillance equipment **22**. Control and network electronics (not shown) within the positioning coupler **10** may also be in communication between the first power and data cable set **36***a* and the motor **24** and the surveillance equipment **22**. The second power and data cable set **36***b* is fed underneath the bearing **28** and pulley **30** and into the internal pathway **16***a* of the column **16** via the lower through-hole **34***a*. The second power and data cable set **36***b* is fed through the internal pathway **16***a* and exits out of the upper through-hole **34***b* and into electrical communication with the surveillance equipment **7**. Advantageously, therefore, the first and second power and data cable sets **36***a*, **36***b* do not obstruct the path of rotation of the arm **18**. That is, the first power and data cable set **36***a* does not elevationally overlap with the arm **18**, but remains underneath the arm **18**, thereby avoiding interference therewith. The second power and data cable set **36***a* elevationally surpasses the arm **18** within the column **16** and exits from the column **16** beyond the arm **18**. Therefore, as shown best in FIG. **5**, the second power and data cable set **36***a* also does not obstruct the path of rotation of the arm **18**. In one configuration, the positioning coupler **10** may also include a slip ring (not shown) to provide communication and power transmission with the surveillance equipment **7** and/or **22** while enabling 360° continuous motion of the surveillance equipment **7** and/or **22**.
[0030] It will be appreciated by those skilled in the art that various modifications and alterations could be made to disclosure above without departing from the broad inventive concepts thereof. Some of these have been discussed above and others will be apparent to those skilled in the art. It is understood, therefore, that this invention is not limited to the particular embodiments disclosed, but it is intended to cover modifications within the spirit and scope of the present disclosure, as set forth in the appended claims.

Claims

1. An equipment mounting assembly comprising: a first plate; a column axially extending from the first plate; an arm rotatably attached to the column and extending beyond a radial extent of the first plate; a surveillance equipment removably mounted to the arm along a portion of the arm outside the radial extent of the first plate; and a motor connected to the column, the motor being (i) rotatable about the column, (ii) positioned outside the radial extent of the first plate, and (iii) configured to rotate the arm about the column.
2. The equipment mounting assembly of claim 1, further comprising a second plate, wherein the column axially extends between the first and second plates.
3. The equipment mounting assembly of claim 2, wherein the arm also extends beyond a radial extent of the second plate
4. The equipment mounting assembly of claim 2, wherein the portion of the arm from which the surveillance equipment projects is also positioned outside a radial extent of the second plate.
5. The equipment mounting assembly of claim 2, wherein the motor is also positioned outside a radial extent of the second plate.
6. The equipment mounting assembly of claim 2, wherein the surveillance equipment is a first surveillance equipment, and wherein the second plate is configured to removably mount a second surveillance equipment thereto.
7. The equipment mounting assembly of claim 2, wherein the surveillance equipment is a first surveillance equipment, and wherein the second plate is configured to mate with a base mounting plate of a second surveillance equipment, the second plate including a clamp configured to removably engage the base mounting plate of the second surveillance equipment in a tool-free

manner, whereby the second surveillance equipment is coaxially mountable upon the assembly.

8. The equipment mounting assembly of claim 1, further comprising a stationary toothed pulley fixedly mounted about the column, a rotatable drive pulley operatively connected to the motor, and a toothed belt extending between the rotatable drive pulley and the stationary toothed pulley.

9. The equipment mounting assembly of claim 8, wherein the drive pulley is positioned outside the radial extent of the first plate.

10. The equipment mounting assembly of claim 1, further comprising a bearing positioned about the column and having an inner race fixedly mounted relative to the column and an outer race rotatably mounted relative to the column, wherein a leading end of the arm is securely attached to the outer race.

11. The equipment mounting assembly of claim 10, wherein the column includes a first through-hole axially underlying the bearing, a second through-hole axially overlying the bearing and an internal pathway within the column and extending between the first and second through-holes.

12. The equipment mounting assembly of claim 1, further comprising the surveillance equipment, wherein the surveillance equipment comprises at least one of a radar, a sensor, a camera, a flood light, a loud hailer, a microphone array, a laser, or another surveillance payload.